Clinical Case Scenario 1



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Clinical Case Scenario

Complications of the Ketogenic Diet

Patient Information:

Name: Sarah

Age: 35

Gender: Female

Medical History: Overweight (BMI:32), diagnosed with type II diabetes mellitus 2 years ago. Managed with metformin (500 mg twice daily).

Social History: Sedentary lifestyle, works as an office assistant, limited physical activity.

Family History: Positive for type II diabetes in mother and maternal grandfather.

Chief Complaint:

Sarah presents to the clinic with complaints of weakness, fatigue, and persistent nausea for the past week.

History of Present Illness:

Sarah reports that she started a ketogenic diet 10 weeks ago to lose weight and better manage her diabetes.

She strictly adhered to a diet high in fats, moderate in proteins, and very low in carbohydrates, consuming less than 20 grams of carbs per day.

Initially, she experienced weight loss and better blood sugar control, but over the past week, she has been feeling progressively worse. She has been experiencing weakness, fatigue, and persistent nausea, despite maintaining her diet regimen.

Physical Examination:

- **Vital Signs:** **BP 130/80 mmHg, HR 88 bpm, RR 16 breaths/min, T 37°C (98.6°F)**
- **General:** **Appears tired and fatigued, overweight.**
- **Cardiovascular:** **Regular rhythm, no murmurs.**
- **Respiratory:** **Clear breath sounds bilaterally.**
- **Abdomen:** **Soft, non-tender,**

Biochemical & hematological Investigations:

Blood Glucose: 120 mg/dL (within target range)

Serum Electrolytes:

- **Sodium (Na^+):** 138 mEq/L
- **Potassium (K^+):** 3.2 mEq/L (low)
- **Chloride (Cl^-):** 101 mEq/L
- **Bicarbonate (HCO^{3-}):** 26 mEq/L (slightly elevated)

Complete Blood Count (CBC): Within normal limits

Liver Function Tests (LFTs): Within normal limits

Urinalysis: Positive for ketones, mild dehydration noted.

Diagnosis:

Complications of the Ketogenic Diet

- **Electrolyte Imbalance (Hypokalemia) and**
- **Metabolic Acidosis**

Note:

Potassium (K^+): **3.2 mEq/L (Low)**

Bicarbonate (HCO^{3-}): **26 mEq/L (Slightly elevated)**

Questions

- What are ketone bodies?
- What is the ketosis & ketogenic (keto) diet and how does it occur?
- What are the primary sources of fuel for the body during ketosis?
- What are the complications and consequences of ketosis?
- Can ketosis lead to nutrient deficiencies? If so, which nutrients are commonly affected?
- How many weeks is considered a safe duration for ketosis?
- What role does hydration play in managing ketosis?
- How does ketosis affect cholesterol levels and cardiovascular health?

What are ketone bodies?

Ketone bodies are water-soluble biomolecules produced by the liver when FAs are broken down for energy in the absence of sufficient glucose.

The three main ketone bodies produced in the human body are:

- **Acetoacetate**
- **Beta-hydroxybutyrate**
- **Acetone**

Notes :

1.Acetoacetate & β-hydroxybutyrate are synthesized from acetyl-CoA, in the mitochondria of liver cells; while acetone is formed by spontaneous decarboxylation of acetoacetate.

2.Its con. in the blood is less than 1 mg/dl.

3.High Con. of ketone bodies can induce ketonemia & ketonuria, & even ketosis & acidosis.

4.Acetoacetate is chemically unstable & breaks down to acetone, which is poorly metabolized, and is excreted in the urine & on the breath.

What is the ketosis & ketogenic (keto) diet and how does it occur?

- **Ketosis is a metabolic state where ketones become an important source of energy for the body & brain,**
- **Unlike FAs, ketones can cross the blood-brain barrier & provide energy for the brain in the absence of glucose.**
- **It occurs when the body shifts from using glucose as its primary fuel source to using fat.**
- **In the early stages of fasting, the use of ketone bodies by the heart & skeletal muscle conserves glucose to support the central nervous system.**
- **With more prolonged starvation, the brain can uptake more ketone bodies to spare glucose consumption**

What is a ketogenic (keto) diet and how does it work?

- **The goal of the keto diet is to shift the body's metabolism away from glucose toward fat metabolism**

The typical macronutrient ratio of a standard ketogenic diet is as follows:

- **70-80% of calories from fat**
- **20-25% of calories from protein**
- **5-10% of calories from carbohydrates**

What are the primary sources of fuel for the body during ketosis?

During ketosis, the primary sources of fuel for the body are FAs derived from dietary fat or stored body fat, which are converted into ketone bodies by the liver.

Formation & utilization of ketone bodies

The last product of FAs in β-oxi. is acetyl CoA, which in many tissues, except the liver, can be completely oxidized to energy, CO_2 & water.

However, in liver cells, a significant amount of acetyl CoA is converted to so-called ketone bodies.

What are the complications and consequences of ketosis?

- **Increased thirst, dry mouth, & Severe dehydration,**
- **Frequent urination,**
- **Fatigue,**
- **Persistent vomiting and abdominal pain,**
- **Confusion,**
- **Difficulty breathing & acetone-like or fruity-smelling breath.**
- **Ketoacidosis**
- **Nutritional deficiencies & electrolyte imbalances**
- **Gastrointestinal issues**
- **Hypoglycemia (low blood sugar)**
- **Increased cholesterol levels**

If these symptoms occur, seek immediate medical attention.

Can ketosis lead to nutrient deficiencies? If so, which nutrients are commonly affected?

Prolonged ketosis can potentially lead to deficiencies in micronutrients such as:

- **Vitamins B & C**
- **As well as minerals like Ca, Mg & K.**

How many weeks is considered a safe duration for ketosis?

For many individuals, a safe duration for ketosis is typically between two to six weeks.

What role does hydration play in managing ketosis?

- **Adequate hydration is essential during ketosis to prevent dehydration and support kidney function.**

- **The body excretes more water and electrolytes while burning fat for fuel.**

How does ketosis affect cholesterol levels and cardiovascular health?

Ketosis may improve some markers of cardiovascular health, such as reducing triglyceride levels and increasing levels of HDL ("good"), but its long-term effects on cholesterol profiles and cardiovascular risk factors are still debated.